

FIT3143 Lab 1

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AI Declaration

I acknowledge the use of ChatGPT (<https://chatgpt.com/>) to assist with the preparation of this assessment. ChatGPT was mainly used to help me understand the lab requirements, explain parallel computing concepts, assist with coding and debugging, plan performance testing, and organise the presentation. All generated code and suggestions were reviewed, tested, and modified by me before submission.

Examples of prompts entered include:

- Give me a basic Task 1 C program for finding prime numbers
- Explain the basic idea of how pthread and OpenMP work in these examples
- How to divide the prime-number workload more evenly between threads
- How to keep the final prime-number output sorted when different threads are doing different parts of the work
- Write a script that runs the program with lots of different values of n and saves the execution times
- Write another test script that runs the pthread and OpenMP versions with different thread counts
- How to organise the runtime and speedup results to make the required graphs
- Give me a rough structure for the presentation slides